Construction of the Inner Product Space F

If $K: \mathcal{X}^2 \to \mathbb{R}$ is symmetric and positive semidefinite, then there is an inner product space $\mathcal{F}$ and a feature map ϕ such that

$$K(\underline{x}, \underline{y}) = \langle \phi_{\underline{x}}, \phi_{\underline{y}} \rangle$$ ← scalar product

$\mathcal{X}$

Let's construct

$$\phi_{\underline{x}}(\underline{z}) := K(\underline{x}, \underline{z})$$ ← feature map ϕ

$$\mathcal{F} := \operatorname{span}(\{\phi_{\underline{x}} : \underline{x} \in \mathcal{X}\})$$ ← space spanned by all linear combinations of the $\phi_{\underline{x}}$ (after completation)

Take two functions from this space

$$f(\underline{z}) = \sum_i \alpha_i \phi_{\underline{x}_i}(\underline{z})$$

$$g(\underline{z}) = \sum_j \beta_j \phi_{\underline{x}_j}(\underline{z})$$

Then:

$$\langle f, g \rangle := \sum_{i,j} \alpha_i \beta_j \langle \phi_{\underline{x}_i}(\underline{z}), \phi_{\underline{x}_j}(\underline{z}) \rangle$$

$$= \sum_{i,j} \alpha_i \beta_j K(\underline{x}_i, \underline{x}_j)$$

is an inner product in $\mathcal{F}$.

① Symmetry: $\langle f, g \rangle \overset{!}{=} \langle g, f \rangle$

$$\langle f, g \rangle = \sum_{i,j} \alpha_i \beta_j K(\underline{x}_i, \underline{x}_j) = \sum_{i,j} \beta_j \alpha_i K(\underline{x}_j, \underline{x}_i) = \langle g, f \rangle$$ symmetric kernel

② Bilinearity:

We make use of the distributivity

$$\langle f, g \rangle = \sum_{i,j} \alpha_i \beta_j K(\underline{x}_i, \underline{x}_j) = \sum_i \alpha_i \sum_j \beta_j K(\underline{x}_i, \underline{x}_j)$$

$$= \sum_i \alpha_i \sum_j \beta_j \phi_{\underline{x}_j}(\underline{x}_i) = \sum_i \alpha_i g(\underline{x}_i)$$

With this we obtain

$$\langle a f^{(1)} + b f^{(2)}, g \rangle = a \sum_i \alpha_i^{(1)} g(\underline{x}_i^{(1)}) + b \sum_j \alpha_j^{(2)} g(\underline{x}_j^{(2)})$$

$$= a \langle f^{(1)}, g \rangle + b \langle f^{(2)}, g \rangle$$

③ positive semi-definite

$$\langle f, f \rangle = \sum_{i,j} \alpha_i \alpha_j K(\underline{x}_i, \underline{x}_j) = \underline{\alpha}^T \underline{K} \underline{\alpha} \geq 0$$ ← pos. semi-definite

$(i, j = 1, \dots, n)$

Comment:

One can always convert a positive semi-definite inner product

to a positive definite one, by taking the quotient space mod.
the subspace of elements $\|\underline{x}\| = 0$.